Sri Lanka Institute of Information Technology



IT3091 – Machine Learning

Year 3, Semester 1 - 2026

Business Problem

EduPredict: Student Performance Prediction
Guided Data Track – Education (Student Performance)

Item	Details
Group	2026-AI-33
Campus	Malabe
Document	Problem framing canvas
Track / Dataset	Guided Data Track (code 3: Education); UCI Student Performance, Mathematics
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Repository	https://github.com/SupunThk/EduPredict

Table P.1: Document submission metadata

Group Members

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Table P.2: Group 2026-AI-33 member details

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1. Project Overview and Scope

1.1 Purpose

We investigate whether student information can estimate final Mathematics grades and help teaching staff identify students who may need academic support. The completed experiment provides grade predictions and review flags; its results do not establish readiness for automatic decisions or early deployment.

1.2 Problem and decision need

Teachers and academic support staff need a way to prioritize students for further review while considering current classroom evidence and student requests. The decision question is what final grade a student may achieve, followed by whether the prediction warrants a supportive conversation. A flag supports staff judgement; it must not determine grades, discipline or access to assistance.

1.3 Task and scope

Element	Project definition
Primary task	Supervised regression predicting G3, the recorded final Mathematics grade.
Unit of analysis	One Mathematics student record; 395 rows and 33 raw columns.
Target	G3 on the recorded 0–20 grade scale; an integer-valued outcome modelled through continuous predictions.
Population	Mathematics records from two Portuguese schools. Retain all 38 G3=0 records, including five in the test set.
Secondary output	Potentially at-risk flag when continuous predicted G3 < 10. Predicted 10 is not flagged.
Primary stakeholder	Teachers and academic support staff; students are affected stakeholders.

Table 1.1: Task and scope definition

2. Prediction Setting and Evaluation

2.1 Information available at prediction time

The main scenario excludes G1 and G2 and uses 30 raw features. The comparison scenario includes both grades and uses 32 raw features; it applies only after the second-period grade is available. The prediction date and availability of the other inputs, particularly absences, remain unresolved. Excluding prior grades alone does not demonstrate early-warning capability.

Predictions remain continuous without post hoc rounding or clipping. A 0–20 target scale does not guarantee that every regression algorithm will output values in that range. The selected forests were checked for out-of-range test predictions in Notebook 04.

2.2 Evaluation and evidence

An 80/20 split with random_state=42 allocates 316 records to training and 79 to final evaluation. Five shuffled training folds compare mean and median baselines, Linear Regression, Ridge, Decision Tree, Random Forest and Gradient Boosting. MAE is the selection metric; RMSE and R^2 support interpretation. Both scenarios selected Random Forest. Accuracy, precision, recall, F1 and specificity describe the derived grade-10 flag, with confusion counts exposing missed students and unnecessary reviews.

Held out predictor	MAE	RMSE	R^2
Training median baseline	3.620	4.534	-0.003
Main: without G1/G2	3.095	3.871	0.269
Later stage: with G1/G2	1.170	1.975	0.810
G2 unchanged (later stage only)	1.000	2.072	0.791

Table 2.1: Held-out test results (79 students)

Of 27 test students with G3 < 10, the main rule found 10 and missed 17, with 11 false positives. The later-stage forest found all 27 with 10 false positives; using G2 < 10 found 26 with six false positives. The G2-unchanged reference also achieved lower MAE than the later-stage forest. Model complexity therefore needs justification against a simple academic reference.

3. Recommendation and Limits

3.1 Success criteria and recommended use

For this study, success means a reproducible comparison, improvement over relevant baselines, transparent error analysis and an honest account of the support tradeoff. Operational acceptance criteria for recall, workload and privacy have not been agreed with a school. The main model's 37.0% test recall is insufficient evidence for relying on its flags alone.

Use the findings to design a staff-reviewed pilot on a new cohort. Verify feature timing, review data errors, and offer support based on staff observations as well as flags. An unflagged prediction must not block assistance. Compare missed students and review workload with a simple G2-based approach at the later stage.

3.2 Limits and evidence location

The small two-school sample limits generalization. Full-dataset EDA preceded splitting, and training validation folds were reused for selection; these limits prevent claiming a fully blind or independent development process. Zero-grade causes are unknown, subgroup checks are descriptive, and predictions do not establish causal effects of interventions.

Evidence: notebooks/03_regression_modelling.ipynb; notebooks/04_evaluation_recommendations.ipynb; Notebook 04 setup and final test tables; Notebook 04 recommendations. Dataset overview: UCI Student Performance, https://archive.ics.uci.edu/dataset/320/student+performance (accessed 12 October 2026).